Chemical Bonding

→ It is a link in form of the net force of attractions existing between atoms/ions in a substance.

→ The resulting substance formed has lower energy than the total energy of separate atoms.

Different types of bonding
↳ Ionic bonding: complete transfer of one or more electrons
↳ Covalent bonding: Sharing of electrons
↳ Metallic bonding: Sea of electrons shared by large number of cations.

Atomic properties determine type of bonding that takes place.

Metal with non-metal: Ionic bonding
↳ metal atom (low IE) loses one or more e^-'s to nonmetal atom (High -ve EA).

Non metal with nonmetal: Covalent Bonding
↳ When atoms differ very little in their ability to lose or gain e^-'s they share electrons

Metal with metal: Metallic bonding
↳ electrons in metallic bonding are delocalized, moving freely through entire piece of metal

Covalent Bonding - Quantum Picture

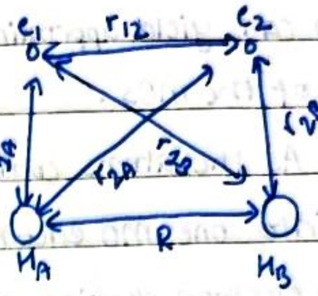
$\hat{H}\psi = E\psi$

$-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi = E\psi$ $\qquad$ $\psi \rightarrow$ ground state wavefunction

We compute distribution of electron density around two nuclei.

$$\hat{H} = \frac{-\hbar^2}{2M}\left(\nabla_A^2 + \nabla_B^2\right) - \frac{\hbar^2}{2m_e}\left(\nabla_1^2 + \nabla_2^2\right)$$

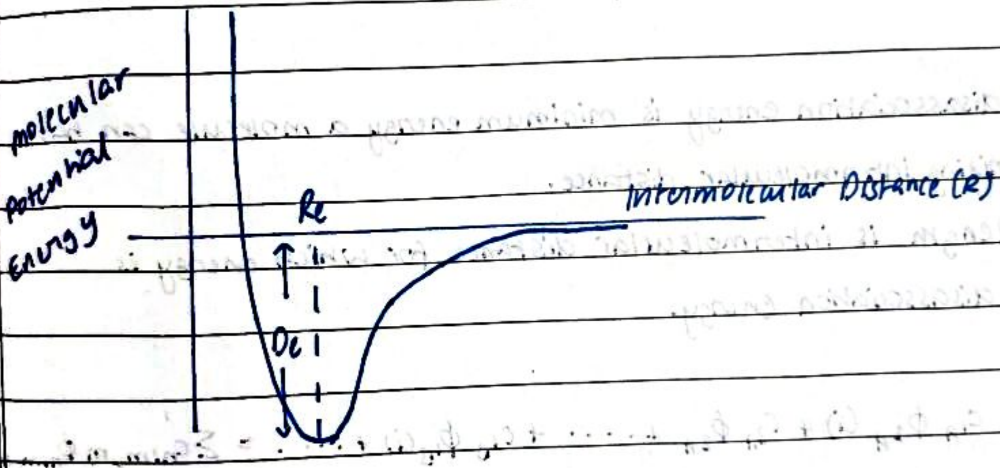
$$\frac{-e^2}{4\pi\varepsilon_0}\left(\frac{1}{r_{1A}} + \frac{1}{r_{1B}} + \frac{1}{r_{2A}} + \frac{1}{r_{2B}}\right) + \frac{e^2}{4\pi\varepsilon_0}\left(\frac{1}{r_{12}} + \frac{1}{R}\right)$$



We apply Born Oppenheimer Approximation

H_2^+

$$\hat{H} = -\frac{\hbar^2}{2m_e}\left(\nabla_1^2 + \nabla_2^2\right) - \frac{e^2}{4\pi\varepsilon_0}\left(\frac{1}{r_{1A}} + \frac{1}{r_{2A}} + \frac{1}{r_{2B}} + \frac{1}{r_{2B}}\right) + \frac{e^2}{4\pi\varepsilon_0 r_{12}}$$



$$\hat{H}\psi_{el} = E_{el}\psi_{el}$$

$$E_{total} = E_{el} + \frac{Z_A Z_B e^2}{4\pi\varepsilon_0 R_{AB}}$$

2 ways to construct molecular electronic wave functions
↳ Linear combination of atomic orbitals (LCAO)
↳ Valence bond (VB)

LCAO – MO
↳ A given molecular orbital is written as a linear combination of atomic orbitals.
↳ generates MO's that are delocalised and builds configurations as per aufbau and is known as molecular orbital Theorem.
→ doesn't predict bondlength, shape of molecule, bond strength
VB → fails to describe some situations, because it ignores wave nature of e^-.
↳ Linear combination used to make hybrid orbitals in VBT involve only valence orbitals on central atom.

MOT

↳ delocalized bonding model.

↳ can yield approximate solutions for set of MOs as well as energies of the MOs.

↳ As the atoms come together two seperate atomic orbitals combine into one MO encompassing both nuclei

↳ electron sharing can only take place if the protons overlap appreciably.

as likelihood an electron will pass through a region of high potential hence depends on how far apart the protons are. If it was 0.15nm proton switches every 10^{-15}s and effective radius of wave function is 0.0536nm so much faster than 10^{-15}s.

Bond disassociation energy is minimum energy a molecule can have on varying intermolecular distance.

Bond length is intermolecular distance for which energy is bond disassociation energy.

$$\psi(i) = c_{1A}\phi_{1A}(i) + c_{2A}\phi_{2A} + \dots + c_{1B}\phi_{1B}(i) + \dots = \sum c_{nlm_l m}\phi_{nlm_l m}(i)$$

molecular wave function expressed as linear combination of AOs.

Prerequisites for choosing atomic orbitals

↳ Energies need to be comparable.

↳ Needs to be able to produce sufficient overlap.

↳ Same symmetry w.r.t interatomic axis.

↳ conservation of orbitals (no of AO = no. of MO)

The square of weighting coefficients gives the probability of finding e^- in that specific AO.

Average value of variable a whose operator is $\hat{a}$ is given by

$$\langle a \rangle = \frac{\int \psi^* \hat{a}\, \psi\, d\tau}{\int \psi^* \psi\, d\tau}$$

If energy is to be found

$$E = \frac{\int \psi^* \hat{H} \psi \, d\tau}{\int \psi^* \psi \, d\tau}$$

$$= \frac{\int (c_1\phi_1 + c_2\phi_2)^* \hat{H} (c_1\phi_1 + c_2\phi_2) \, d\tau}{\int (c_1\phi_1 + c_2\phi_2)^* (c_1\phi_1 + c_2\phi_2)}$$

$$= \frac{c_1^2 \int \phi_1^* \hat{H} \phi_1 \, d\tau + 2c_1c_2 \int \phi_1^* \hat{H} \phi_2 \, d\tau + c_2^2 \int \phi_2^* \hat{H} \phi_2 \, d\tau}{c_1^2 \int \phi_1^* \phi_1 \, d\tau + 2c_1c_2 \int \phi_1^* \phi_2 \, d\tau + c_2^2 \int \phi_2^* \phi_2 \, d\tau}$$

$$= \frac{c_1^2 H_{11} + 2c_1c_2 H_{12} + c_2^2 H_{22}}{c_1^2 S_{11} + 2c_1c_2 S_{12} + c_2^2 S_{22}}$$

where $H_{ij} = \int \phi_i^* \hat{H} \phi_j \, d\tau$ and $S_{ij} = \int \phi_i^* \phi_j \, d\tau$

Also $\int \phi_1^* \hat{H} \phi_2 \, d\tau = \int \phi_2^* \hat{H} \phi_1 \, d\tau$ (Hermitian Property)

To minimize the energy of trial wavefunction we must differentiate them w.r.t c_1 and c_2 and set that to 0.

We get

$$c_1(H_{11} - ES_{11}) + c_2(H_{12} - ES_{12}) = 0 \quad - (1)$$

$$c_1(H_{12} - ES_{12}) + c_2(H_{22} - ES_{22}) = 0 \quad - (2)$$

$$\begin{vmatrix} H_{11} - ES_{11} & H_{12} - ES_{12} \\ H_{12} - ES_{12} & H_{22} - ES_{22} \end{vmatrix} = 0$$

$S_{11} = \int 1s_A^* \, 1s_A \, d\tau = 1$

$S_{22} = \int 2s_A^* \, 2s_A \, d\tau = 1$

S_{12} is the overlap integral.

$$S_{12} = \int 1s_A^* \, 1s_B \, d\tau$$

$S_{12} \neq 0$ whenever two s-orbitals (of same phase) are combined or if s-orbital p-orbital overlap then iff there is net constructive -interference.

$S_{ij} = S_{AB}(R) = \left(1 + R + \frac{R^2}{3}\right)e^{-R}$

For homonuclear diatomic molecule H_2 the 2 AOs wavefunctions are identical

$\therefore H_{11} = \int 1s_A^* \hat{H}\, 1s_B\, d\tau = \int 1s_B^* \hat{H}\, 1s_B\, d\tau = H_{22}$

$$\begin{vmatrix} H_{11} - E & H_{12} - ES_{12} \\ H_{12} - ES_{12} & H_{11} - E \end{vmatrix} = 0$$

$$(H_{11} - E)^2 - (H_{12} - ES_{12})^2 = 0$$

$$H_{11} - E = H_{12} - ES_{12} \quad \text{(or)} \quad H_{11} - E = ES_{12} - H_{12}$$

$$\frac{H_{11} - H_{12}}{1 - S_{12}} = E \quad \text{(or)} \quad E = \frac{H_{11} + H_{12}}{1 + S_{12}}$$

Energy of antibonding MO Energy of bonding MO.

Substituting E_{BMO} in ① and ② we get $c_1/c_2 = 1$

$\psi_b = c_1\, 1sA + c_1\, 1sB = c_1(1s_A + 1s_B)$

By normalisation

$$\int (\psi_b)^2 d\tau = 1$$

$$c_1^2 \int (1sA + 1sB)^2 d\tau = 1$$

$$c_1^2 \left(\int (1sA)^2 d\tau + 2\int 1sA\, 1sB\, d\tau + \int (1sB)^2 d\tau \right) = 1$$

$$c_1^2 (1 + 1 + 2S) = 1$$

$$c_1 = \frac{1}{2+2S} \Rightarrow c_1 = \frac{1}{\sqrt{2(1+S)}}$$

similarly we find $c_1/c_2 = -1$ if we put $E = E_{ABMO}$

and $c_2 = \frac{1}{\sqrt{2(1-S)}}$

$$\overset{\sigma_{1s}^*}{\psi_{abmo}} = \frac{1sA - 1sB}{\sqrt{2(1-S)}} \qquad \overset{\sigma_{1s}}{\psi_{Bmo}} = \frac{1sA + 1sB}{\sqrt{2(1+S)}}$$

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$\psi_{BMO} \propto (2s_A + 2s_B) \propto (e^{-kr_A} + e^{-kr_B})$

$\psi_{ABMO} \propto (2s_A - 2s_B) \propto (e^{-kr_A} - e^{-kr_B})$

Shifting e^- from away from nucleus into internuclear region would raise its potential energy however according to our formulas it should decrease.

This is explained by the fact that atomic orbitals shrink and it brings e^-'s closer to the nuclei and the following increase in attraction is greater than potential energy loss moving into the centre.

In H_2^+

$$\hat{H}_{e1} = \frac{-\hbar^2}{2m_e}(\nabla_1^2) - \frac{e^2}{4\pi\varepsilon_0}\left(\frac{1}{r_{A1}} + \frac{1}{r_{B1}}\right)$$

in a.u: $-\nabla_1^2 - \frac{1}{r_A} - \frac{1}{r_B}$

$$H_{11} = \int 2s_A^* \left(-\nabla^2 - \frac{1}{r_{A1}} - \frac{1}{r_{B1}}\right) 2s_A \, d\tau$$

$2s_A^* = 2s_A$

$\phi_{1s} = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0}$

$$= \int 2s_A \left(-\nabla^2 - \frac{1}{r_{A1}}\right) 2s_A \, d\tau - \int 2s_A \left(\frac{1}{r_{B1}}\right) 2s_A \, d\tau$$

$$E_{H_{2s}} = \int 2s_A \left(-\nabla^2 - \frac{1}{r_{A1}}\right) 2s_A \, d\tau$$

Coloumb Integral $J = \int 2s_A \left(\frac{1}{r_{B1}}\right) 2s_A \, d\tau$

$H_{11} = E_{H_{2s}} - J$

Now for

$$H_{12} = \int 2s_A \left(-\nabla^2 - \frac{1}{r_{A1}} - \frac{1}{r_{B1}}\right) 2s_B \, d\tau$$

$$= \int 2s_A \left(-\nabla^2 - \frac{1}{r_{A1}}\right) 2s_B \, d\tau - \int 2s_A \left(\frac{1}{r_{B1}}\right) 2s_B \, d\tau$$

$2s_B$ is eigen function of operator with eigen value $E_{H_{2s}}$

Exchange Integral (K)

$= E_{H_{2s}} S - K$ (electron is in one orbital on one side of operator and in different orbital on other side)

$J \rightarrow$ attraction of proton B due to electron density of H_A

$K \rightarrow$ attraction of proton due to build up of electron charge density between two protons.

$$E_B = \frac{H_{11} + H_{12}}{1+S} = \frac{E_{H_{1s}}(1+S) - J - K}{1+S} = E_{H_{1s}} - \frac{(J+K)}{1+S}$$

$$E_{ABMO} = \frac{H_{11} - H_{12}}{1+S} = \frac{E_{H_{1s}}(1-S) - J + K}{1-S} = E_{H_{1s}} - \frac{(J-K)}{1-S}$$

$$\Delta E_b \,(\text{w.r.t } E_{H_{1s}}) = -\frac{J+K}{1+S} = E_b - E_{H_{1s}}$$

$$\Delta E_a = \frac{-(J-K)}{1-S}$$

$1-S > 0$

$J < K$

$$\Delta E_a > \Delta E_b$$

Bond energy = (number of e^- in BMO) $\times$ (ΔE_b).

Now H_2 has $2e^-$'s. We have one bonding wavefunction $\psi_{\sigma_{1s}}(1)$ ~~and~~ for 1st e^- and $\psi_{\sigma_{1s}}(2)$ for 2nd e^-.

The bonding molecular wavefunction for H_2 is

$$\psi^{+}_{\sigma_{1s}} = \psi_{\sigma_{1s}}(1)\, \psi_{\sigma_{1s}}(2) = \left[\frac{1s_A(1) + 1s_B(1)}{\sqrt{2(1+S)}}\right]\left[\frac{1s_A(2) + 1s_B(2)}{\sqrt{2(1+S)}}\right]$$

$$= \left[\frac{1s_A(1)\,1s_B(2) + 1s_A(2)\,1s_B(2) + 1s_A(1)\,1s_A(2) + 1s_B(1)\,1s_B(2)}{2(1+S)}\right]$$

According to Pauli's principle, the complete wavefunction must be antisymmetric.

$\psi^{+}_{\sigma_{1s}}$ is symmetric and hence should be multiplied by antisymmetric spin factor $\sigma^{spin}_{-} = \frac{1}{\sqrt{2}}[\alpha(1)\beta(2) - \beta(1)\alpha(2)]$

$$\sigma_{1s} = \psi^{+}_{1s}\, \sigma^{spin}_{-}$$

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Valence Bond

↳ As atoms come closer so that bond formation takes place we don't know whether e_1^- came from proton A and e_2^- came from proton B or vice versa. So we can propose a approximate wave function as a linear combination ψ_1 and ψ_2

$\psi_1 = \psi_A(1)\psi_B(2) \qquad \psi_2 = \psi_B(1)\psi_A(2)$

$\psi_{covalent} = c_1\psi_1 + c_2\psi_2$

$H_{11} = \int(\psi_A(1)\psi_B(2))\,\hat{H}\,(\psi_A(1)\psi_B(2))\,d\tau$

$E_+ = \dfrac{H_{11}+H_{12}}{1+S} \qquad E_- = \dfrac{H_{11}-H_{12}}{1-S}$

$S_{12} = S^2$ where $S \rightarrow$ overlap integral

$\psi_+ = \dfrac{1}{\sqrt{2(1+S^2)}}[\psi_A(1)\psi_B(2)+\psi_B(1)\psi_A(2)]$

$\psi_- = \dfrac{1}{\sqrt{2(1-S^2)}}[\psi_A(1)\psi_B(2)-\psi_A(2)\psi_B(1)]$

So complete wavefunction

$\psi_+(spin) = \dfrac{1}{\sqrt{2(1+S^2)}}[\psi_A(1)\psi_B(2)+\psi_B(1)\psi_A(2)]\dfrac{1}{\sqrt{2}}[\alpha(1)\beta(2)-\alpha(2)\beta(1)]$

$\psi_{ionic} = \psi_A(1)\psi_A(2)+\psi_B(2)\psi_B(1)$

$\psi_{AB} = \psi_{covalent} + \lambda\psi_{ionic}$

↳ quantifies ionic character.